

COMPUTER PROGRAMMING

LAB REPORT 5

TITLE:

Using different loops

1. Objective

- The main goal of this lab is to practice using different types of loops in C++. Loops are one of the most important concepts in programming because they allow us to repeat tasks without writing the same code again and again.

2. Theory

- This lab focuses on while loops, nested while loops, and do...while loops. These loops help in controlling repetition in programming tasks efficiently.

2.1 While Loop

- The while loop repeats a block of code as long as a condition remains true. The condition is checked before execution of the loop body.

2.2 Nested While Loop

- A nested loop is a loop placed inside another loop. It is commonly used for patterns, tables, and repeated structures.

2.3 Do...While Loop

- The do...while loop executes the body at least once before checking the condition.

3. Programs and Outputs

Task 1: Print numbers 1 to 10 using a while loop.

Task 2: Repeat Task 1 using a do...while loop.

Task 3: Calculate the sum of numbers from 1 to n using a while loop.

4. Discussion

- This lab demonstrated how loops simplify programming tasks by reducing repetitive code. The while loop is useful when conditions control execution, while the do...while loop ensures at least one execution. Nested loops help in pattern generation and multi-level repetition.

5. Conclusion

- This lab improved understanding of loops and their practical applications in programming. It also helped in learning how to control program flow effectively.

Sample Code Example

```
#include <iostream>
using namespace std;

int main() {
    int i = 1;
    while (i <= 10) {
        cout << i << endl;
        i++;
    }
    return 0;
}
```